Surveying Solved Problems

Fifth Edition

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for the NCEES
FS and PS Surveying Exams

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Preface

Over the past 40 years, I have seen the everyday work of surveyors change from transit and tape to the global positioning system (GPS). Calculation tools have evolved from logarithmic tables to personal computers. In surveying today, you have to run as fast as you can to keep up with new methods and technologies. As technology grows more complex, the requirements for licensure become more challenging. As a result, it is imperative that you spend the time to adequately prepare for your exam.

The problems in *Surveying Solved Problems* cover a broad range of the surveying methods and knowledge areas essential for modern practice. They are from the "real world" of surveying. From total stations to GPS, and taping corrections to hydrography, the topics included in this book are broad.

For this fifth edition, I have written new problems covering the following topics: land development; BLM *Manual of Surveying Instructions*; FEMA requirements; risk management procedures; ALTA/NSPS Land Title Surveys; route surveys for alignments and utilities; surveys to establish new parcels, lots, or units; as-built/record drawing surveys; consultation services.

It is my sincere hope that you will find this book helpful, whether you are a student or a seasoned surveyor.

Jan Van Sickle, PhD, PLS

Introduction

ABOUT THIS BOOK

Surveying Solved Problems is designed to prepare you for the National Council of Examiners for Engineering and Surveying (NCEES) Fundamentals of Surveying (FS) exam and the Principles and Practice of Surveying (PS) exam. The included problems cover a broad range of subject areas relevant to surveying. These problems are both conceptual and practical, and they are representative of the type and difficulty of the problems you will encounter on the FS and PS exams.

Problems are grouped by topic into 17 chapters. Each chapter is further organized by subtopics covering many of the FS and PS exam specifications. Within each subtopic, problems are ordered from easiest to most challenging. Like the actual exam, each of the problems has four answer options, labeled (A), (B), (C), and (D). If the options are numerical, they will be displayed in increasing value. One of the answer options is correct (or, will be "most nearly" correct). The remaining answer options are incorrect and may consist of one or more "logical distractors," the term used by NCEES to designate incorrect options that result from simple mathematical errors, such as failing to square a term in an equation, or more serious errors, such as using the wrong equation.

The quantitative solutions in this book are presented step-by-step to help you follow the logical development of the solving approach and to provide examples of how you may want to solve similar problems on the exam. The qualitative solutions explain the correct answer and present related supportive information, such as surveying laws, regulations, and history.

HOW TO USE THIS BOOK

The chapters may be worked in any order, but it is recommended that you work only one chapter at a time. Problems within a chapter are categorized by subtopic, and they should be worked in order as they proceed from the simplest to the most difficult. Before attempting to solve problems, review the topics covered and identify those topics with which you're less familiar. Then, work a few of these problems to assess your general understanding, and identify your strengths and weaknesses. Try to solve problems without looking at the solutions in this book. Then, you can use the solutions to check your work or for guidance in solving problems you found difficult. Every step needed to arrive at the final answer, whether it be a function of logic or of calculation (and sometimes both), is provided so that you can determine which topics need further study.

Once you've identified which topics will require more study, locate relevant resource materials (such as those provided in the previous section), and work the problems in one subject area at a time. You should also work problems using the appropriate NCEES *Handbook*, as it will be the only resource you'll be able to use during your exam.

After you have worked all the problems with which you're least familiar, move on to the problems whose subject areas you are more familiar with. Follow the same procedure as previously described until you have worked all problems.

Remember that the value of a collection of problems such as this does not lie in its ability to guide your preparation. Rather, the value is in giving you an opportunity to bring together all your knowledge and to practice your problem-solving skills.

THE FS AND PS EXAMS

The FS and PS exams are standardized tests prepared by the NCEES to ensure that only qualified people are legally allowed to practice as surveyors. To ensure the reliability and validity of the tests, the FS and PS exams are based on input from committees of professional surveyors and educators throughout the United States.

The FS exam tests general entry-level surveying principles you are expected to have gained through academic study. In most states, passing the FS exam is a requirement for registration as a surveyor-in-training or surveyor intern. The PS exam tests your ability to apply those principles to the kinds of problems typically encountered in professional practice. Passing the PS exam is required in most states for full licensure as a professional surveyor. This may be necessary if your company requires licensure for employment or advancement, if your state requires registration before you may use the title "Surveyor," if you wish to be an independent consultant, or for other reasons. The PS exam is usually administered in conjunction with an exam on surveying practices and regulations specific to the state administering the exam.

Although both the FS and PS exams are prepared by NCEES, they are administered under the direction of the licensing boards of the various states, which usually also require you to have attained a certain level of education and (for the PS exam) experience before taking the exams. For information regarding these requirements in the state in which you plan to become licensed, and to apply for licensure, contact that state's board. Current addresses and phone numbers for each state board may be obtained at PPI's website, **ppi2pass.com**.

THE FS EXAM

Structure

The FS exam is a computer-based test that concentrates on the fundamentals and basics of surveying. You may take it at any Pearson VUE test center.

The exam contains 110 problems. Only one problem is given onscreen at a time. The exam is not adaptive (i.e., your response to one problem has no bearing on the next problem you are given); even if you answer the first five mathematics problems correctly, you'll still have to answer the sixth problem.

Your exam will include a limited (and unknown) number of problems that will not be scored and will not have an impact on your results (known as "pretest items"). NCEES does this to evaluate potential problems for future exams. You won't know which problems are pretest items. They are not identifiable and are randomly distributed throughout the exam.

The FS exam is 6 hours long and includes an 8-minute tutorial, a 25-minute break, and 2 minutes to sign a nondisclosure agreement. The total time you'll have to answer the exam problems is 5 hours and 20 minutes, which works out to slightly less than 3 minutes per problem. However, the exam does not pace you. You may spend as much time as you like on each question. You may can take less than a 25-minute break, if you like, but you cannot work through the break, and the break time cannot be added to the time you have to complete your exam. You may also leave your seat for personal reasons, but the "clock" does not stop for your absence.

You may work through the problems in any sequence. If you want to go back and check your answers before you submit them for grading, you may. However, once you submit your answers, you are not able to go back and review them. Unanswered problems are scored the same as problems answered incorrectly, so you should use the last few minutes of your exam time to guess at all unanswered problems.

The NCEES Nondisclosure Agreement

At the beginning of the FS exam, a nondisclosure agreement will appear on the screen. To begin the exam, you must accept the agreement within 2 minutes. If you do not accept within 2 minutes, your exam appointment will end, and you will forfeit your appointment and exam fees. The nondisclosure agreement is discussed in the section titled "Subversion After the Exam." The nondisclosure agreement, as stated in the *NCEES Examinee Guide*, is as follows.

> This exam is confidential and secure, owned and copyrighted by NCEES, and protected by the laws of the United States and elsewhere. It is made available to you, the examinee, solely for valid assessment and licensing purposes. To take this exam, you must agree not to disclose, publish, reproduce, or transmit this exam, in whole or in part, in any form or by any means, oral or written, electronic or mechanical, for any purpose, without the prior express written permission of NCEES. This includes agreeing not to post or disclose any test questions or answers from this exam, in whole or in part, on any websites, online forums, or chat rooms, or in any other electronic transmissions, at any time.

Your Exam Is Unique

The exam that you take will not be exactly the same exam taken by the person sitting next to you. NCEES says that, for each examinee, its computer-based testing (CBT) system randomly selects different but equivalent problems from its database using a linear, on-the-fly (LOFT) algorithm. Each examinee will have a unique exam, and all exams will be of equivalent difficulty.

The Exam Interface

The onscreen exam interface contains only minimal navigational tools. Onscreen navigation is limited to selecting an answer, advancing to the next problem, going back to the previous problem, and flagging the current problem for later review. The interface also includes a timer, the current problem number (e.g., 45 of 110), a pop-up scientific calculator, and access to an onscreen version of the *FS Reference Handbook*.

During the exam, you can advance through the problems in sequence, but you cannot jump to any specific problem, whether or not it has been flagged. After you have completed the last problem in a session, however, the navigation capabilities change, and you are permitted to review problems in any sequence and navigate to flagged problems.

Knowledge Areas and Problem Distribution

The FS exam includes problems in 13 knowledge areas. Each area is listed as follows, with the number of problems relating to this area that you can expect to see on your exam and a list of topics included in this area.

1. **Mathematics (13–20 problems):** algebra, trigonometry, and basic geometry; spherical trigonometry; linear algebra and matrix theory; analytic geometry and calculus

2. **Basic sciences (5–8 problems):** geology, dendrology, cartography, and environmental sciences
3. **Spatial data acquisition and reduction (6–9 problems):** vertical measurement, distance measurement, angle measurement, unit conversions, redundancy, knowledge and utilization of instruments and methods, and understanding of historical methods and instruments
4. **Survey computations and computer applications (19–29 problems):** coordinate geometry, traverse closure and adjustment, area, volume, horizontal and vertical curves, spirals, and spreadsheets
5. **Statistics and adjustments (6–9 problems):** mean, median, and mode; variance and standard deviation; error analysis; least squares adjustment; measurement and positional tolerance; and relative, network, and positional accuracy
6. **Geodesy (5–8 problems):** basic theory, satellite positioning, gravity, coordinate systems, datums, and map projections
7. **Boundary and cadastral survey law (13–20 problems):** controlling elements, gathering and identifying evidence, records research, legal descriptions, case law, riparian rights, public land survey system, metes and bounds, simultaneously created parcels, easements and encumbrances
8. **Photogrammetry and remote sensing (4–6 problems):** interpretation and analysis, project and flight planning, quality control, ground control, and LiDAR
9. **Survey processes and methods (11–17 problems):** land development (principles, standards, and regulations); boundary location; mapping, cartography, and topography; construction; riparian surveys; route surveying; and control surveys
10. **Geographic information systems (GIS) (5–8 problems):** feature collection and integration, database concepts and design, accuracy and use, and metadata
11. **Graphical communication and mapping (6–9 problems):** plans and specifications; contours and slopes; scales; planimetric features and symbols; land forms; digital terrain modeling and digital elevation modeling; and survey maps, plats, drawings, and reports
12. **Professional communication (4–6 problems):** oral and written communication; alternative forms of communication; documentation and record-keeping
13. **Business concepts (3–5 problems):** contracts, liability and risk management, financial practices, leadership and management principles, personnel management principles, project planning and design, ethics, and safety

Exam content is subject to change. Consult PPI's website (**ppi2pass.com**) for current specifications.

THE PS EXAM

Structure

The PS exam is also a computer-based, closed-book exam. There are 100 problems on the exam. A PS testing session is 7 hours long, which includes 2 minutes for completion of a nondisclosure agreement, 8 minutes for a tutorial, and 6 hours for actually taking the exam. The exam is divided into two sections, each with approximately 50 problems, and an optional 50-minute scheduled break between the sections.

Knowledge Areas and Problem Distribution

The PS exam includes problems in five knowledge areas. Those areas are listed here with their subtopics and the number of problems to expect in each knowledge area.

1. Legal principles (18–27 problems)

Principles of Evidence: how to search for data and for physical evidence to evaluate data; how to evaluate data; parol evidence; prescriptive rights; adverse possession; acquiescence; controlling elements; easement rights

Common Law Boundary Principles: historical and current common law principles; riparian and littoral rights; sovereign rights, including both navigable waters and eminent domain; sovereign land grants

Sequential and Simultaneous Conveyance Concepts: types of conveyances; junior/senior rights; record and physical evidence

Legal Descriptions for Real Property Transactions: preparation and interpretation of legal descriptions; controlling elements and how they impact the description; unwritten rights and how they impact the description; encumbrances and how they impact the description; easements and how they impact the description

Evidence for the Perpetuation of the U.S. PLSS

2. Professional survey practices (22–33 problems)

Public/Private Record Sources: resources for private and public records; local public records indexing and filing system; local survey office records

Documentation, Supervision, and Clear Communication of Field Procedures: field surveying techniques; field surveying practices; data collection protocols

GPS/GNSS Including Satellite Constellations, Static GPS, RTK, PPP, and Virtual Networks

Surveying Principles and Computations: technical computations; applicable software

Monumentation Standards: applicable monumentation criteria; monument types

Land Development Solutions: regulatory land development criteria; construction criteria; land development implementation procedures

Survey Maps/Plats/Reports: technical communications by schematic, platting, and mapping processes and procedures; communication options

GIS: GIS spatial databases and metadata; datums and projections related to GIS

3. Standards and specifications (8–12 problems)

BLM Manual of Surveying Instructions

ALTA/NSPS Land Title Survey Standards: current ALTA/NSPS Land Title Survey Standards; state statutes regarding boundary surveys in conjunction with ALTA/NSPS Land Title Surveys

FEMA Requirements: FEMA specifications and instructions; horizontal and vertical datums related to FEMA flood zones; current FEMA elevation certificate; FEMA Flood Insurance Study

4. Business practices (13–19 problems)

General Business Practices and Procedures: project planning and project management; deliverables; costs, budgets, and contracts; types of surveys; site features and conditions; scope of services; appropriate equipment and instruments

Risk Management Procedures: safety procedures; QA/QC methods; risk management in contracts; insurance needs and requirements; potential liabilities

Professional Conduct

Communication with Clients, Staff, Related Professions, and the Public: different forms of communications; appropriate type of communication to convey concepts; related professions and their impact on client needs and deliverables

5. Areas of practice (24–36 problems)

ALTA/NSPS Land Title Surveys: legal documents, such as deeds, easements, and agreements; zoning information as applied to ALTA/NSPS Land Title Surveys; title insurance commitment letters and policies; underground features as applied to ALTA/NSPS Land Title Surveys

Control Networks and Geodetic Network Surveys: datums and reference frames relative to control networks; differences between local datums and geodetic datums; equipment appropriate for control surveys; the Federal Geographic Data Committee Geospatial Positioning Accuracy Standards; the National Geospatial Programs (NGP) Standards and Specifications—Digital Data Standards

Construction Surveys: construction plan reading; construction calculations including slopes, grades, and plan details; construction techniques and activities; horizontal and vertical positioning relative to a plan or datum

Boundary Surveys: physical boundary evidence; boundary reconciliations; historical measurement accuracy, equipment, and techniques; legal principles related to boundary surveys

Route Surveys for Alignments and Utilities: route alignment stationing practices; reading and interpreting roadway and utility plans

Topographic: topographic/planimetric mapping and control standards; interpretation, reconciliation, and adjustment of topographic survey data; QA/QC procedures as applied to topographic surveys; ground, hydrographic, and remote sensing equipment; the U.S. National Map Accuracy Standards as applied to topographic surveys; tools and techniques required to perform hydrographic, bathymetric, and remote sensing surveys; nomenclature related to utilities

Surveys to Establish New Parcels, Lots, or Units: types of subdivisions; platting; condominiums and associations; deed restrictions and restrictive covenants; zoning and subdivision ordinances

As-Built/Record Drawing Surveys: as-built/record drawing calculations including slopes, grades, and plan details; as-built/record drawing techniques and activities; horizontal and vertical as-built/record drawing positions relative to a plan or datum

Consultation Services: site topography and slope for development purposes; site access for development purposes; zoning standards related to new projects; floodplains as related to land development

Note: Exam content is subject to change. Consult PPI's website (**ppi2pass.com**) for current specifications.

TYPICAL PROBLEM FORMAT FOR THE FS AND PS EXAMS

The multiple-choice problems on the FS exam are typically short, straightforward, and designed to test your knowledge of the fundamentals of surveying and mapping.

The PS exam is designed to test your ability to apply surveying fundamentals to typical problems encountered in surveying practice. For example, a series of land descriptions from deeds might be provided, followed by problems requiring you to analyze the descriptions, establish certain boundaries or corner positions, and treat encroachments.

Most of the problems in both exams will be in the traditional multiple-choice exam, with a problem statement followed by four answer options. But, with the new CBT format, the exams may include alternative item type (AIT) problems, such as multiple correct problems, fill-